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Transportation infrastructure and college admissions quality: Evidence from China's National College Entrance Examination

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ABSTRACT

We exploit the unique context of China's National College Entrance Examination (NCEE) to investigate the impact of transportation infrastructure on college admissions quality. Our analysis focuses on China's rapidly expanding high-speed rail (HSR) network. We provide empirical evidence that the opening of HSR significantly improves the quality of college admissions, as reflected in higher admission scores. We examine three potential mechanisms: (1) HSR promotes local economic development, (2) HSR enhances college strength, and (3) HSR improves transportation accessibility. We present evidence suggesting that the first two mechanisms cannot account for our main findings, whereas the third mechanism can. Moreover, our results imply that HSR may widen regional disparities in college admissions quality: Colleges in developed regions benefit from HSR by attracting higher-quality students from developing regions, while colleges in those developing regions do not experience similar gains. Using data from the Chinese population census, we further corroborate this finding by showing that, relative to developing regions, developed regions attract a larger inflow of highly educated individuals from HSR. Within the context of China's higher education in this paper, these findings provide new empirical evidence on how HSR affects college admissions quality, talent mobility and the spatial distribution of talent.

1. Introduction

Over the past several decades, global higher education has undergone rapid expansion (Marginson, 2016). As of now, approximately 254 million students are enrolled in universities worldwide—more than double the number from two decades ago (UNESCO, 2025). Correspondingly, there are around 50,000 universities globally (Uhomes, 2025). A growing body of news reports, academic studies, and anecdotal evidence indicates that competition among universities for student admission has intensified. Universities increasingly strive to attract higher-aptitude students in order to maintain or improve their rankings, reputations, and overall status (Killgore, 2009). To this end, universities invest substantial resources in recruitment efforts, such as offering more generous financial aid packages, lowering tuition fees, and constructing state-of-the-art facilities (Alter and Reback, 2014; Bauld, 2023). Similarly, student-side competition has become fierce, as applicants seek admission to more prestigious institutions. In the United States, for instance, college admissions have become significantly more competitive over the past few decades (Bound et al., 2009). Ivy League universities have seen declining acceptance rates year after year, and their intense selectivity has become widely recognized (Powell, 2016; Ma, 2024). This competitive pressure has also extended to institutions that are less elite but are still plenty competitive (Jaschik, 2022). Given this context, there is growing scholarly and public interest in the characteristics that enhance a university's ability to attract high-aptitude students. However, relatively less attention has been paid to how external environmental factors influence a university's attractiveness and admission quality. Universities typically recruit on a national scale, implying that geographic distance and commuting costs are critical considerations in student decision-making. Transportation infrastructure, therefore, plays a non-negligible role. In this study, we leverage the distinctive context of China's higher education system and high-speed rail (HSR) to provide new empirical insights into how transportation infrastructure affects the quality of college admission.

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0166-0462/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

China offers an ideal empirical setting for the present study. First, it possesses one of the largest higher education systems in the world, with approximately 3000 universities and colleges, and around 10 million new college entrants each year (Eble and Hu, 2025). Second, since 2008, China has experienced an unprecedented expansion of its HSR network, which has rapidly developed into the world's largest and most extensively used HSR system within just over a decade (Fang et al., 2025). Our analysis centers on China's National College Entrance Examination (NCEE), which provides a particularly suitable context for examining college admission. The NCEE has several defining features that make it well-suited for this study. First, the NCEE is massive in scale. Fig. 1 presents the annual number of examinees, admitted students, and admission rates from 2006 to 2018—the period covered by this study. As shown, the number of NCEE examinees remained relatively stable at around 10 million per year, while the number of admitted students steadily increased. The admission rate rose from under 60% in 2006 to over 80% in 2018. Second, with the exception of a few small-scale special admissions programs, Chinese universities and colleges do not have autonomous admissions rights. The vast majority of enrolled students are admitted through the NCEE. Third, the NCEE is a nationally standardized examination organized by the central government—this is also reflected in its formal title: the "Nationwide Unified Examination for Admissions to General Universities and Colleges." The exam is administered under a uniform and highly regulated evaluation system across all candidates (for details, see Section 2.1). These institutional features ensure, first, that the NCEE is representative in terms of scale, and second, that the NCEE scores are a reliable indicator of student aptitude (Eble and Hu, 2025). Accordingly, we use the admission score of colleges as a proxy for admission quality—higher scores reflect a higher aptitude of admitted students.

We constructed the dataset required for our empirical analysis. Specifically, we obtained data on college admission scores from 2006 to 2018 from Sina Gaokao (https://edu.sina.com.cn/gaokao/), a long-established and comprehensive information platform for NCEE in China. This dataset includes average and maximum admission scores at the level of college-year- admission province (the provinces from which colleges recruit students)-track (humanities or science)-admission batch (Batch 1, Batch 2, and Batch 3 undergraduate programs). Our focus is on undergraduate admissions under the NCEE system. As such, the institutions analyzed in this study are generally closer in nature to universities rather than colleges. To maintain consistency with NCEE terminology, we use the term college instead of university throughout this paper. We identified a unique institutional identifier for each college and collected college-level data on personnel, funding, research projects, and scientific achievements. In addition, we obtained data on HSR from the Chinese Research Data Services Platform (CNRDS) database to determine the timing of HSR openings at the city[1] level. Detailed information on the data sources and construction is provided in Section 3.1.

Based on the aforementioned dataset and a difference-in-differences (DID) approach, our empirical analysis finds that HSR significantly improves the quality of college admissions. This effect is reflected in a statistically significant increase in the average admission scores of colleges following the opening of HSR in their host cities. According to our baseline regression results, the opening of HSR leads to a 0.372 percentage point increase in the national percentile ranking of colleges' average admission scores. This qualitative finding also holds for the maximum admission scores, suggesting that HSR not only enhances the overall quality of admitted students but also improves the quality of the top-performing entrants. Given the large scale and highly competitive nature of the NCEE, the effect identified in this study is substantial and should not be overlooked. A more detailed discussion of these results is provided in Section 4.1.

We posit that, within the context of this study, there are three potential mechanisms through which HSR improves the quality of college admissions. First, HSR promotes city economic development (Potential Mechanism 1). Second, HSR enhances the overall strength of colleges (Potential Mechanism 2). Third, HSR improves transportation convenience (Potential Mechanism 3). Theoretically, colleges can attract higher-aptitude students through any of these three mechanisms. We empirically examine each of these potential channels. To investigate Potential Mechanisms 1 and 2, We adopt two approaches. First, we adopt the approach of Chor et al. (2021) and Dippel et al. (2022), conducting mechanism analysis by controlling for potential mechanism variables within the baseline regressions.[2] Second, we empirically examine the effects of HSR on the mechanism variables. Our findings suggest that Potential Mechanism 1 and 2 do not account for the core effect identified in this paper. Regarding Potential Mechanism 3, we provide evidence in support of Potential Mechanism 3. We find that the effect of HSR on raising college admission scores is stronger when the HSR station is located in the city center or when the city plays a central role in the HSR network (e.g., serving as a intersection or departure or terminal). More importantly, we construct a bilateral HSR connectivity measure between the college city and the admission province, and estimate a gravity model. The empirical results show that bilateral HSR connectivity significantly raises colleges' admission scores. This finding highlights the role of HSR in improving transportation convenience and reducing bilateral commuting time and costs.

In the final part of our empirical analysis, we explore the heterogeneous effects of HSR on college admissions quality across different regions in China. The "Great Migration" in China represents one of the largest human and labor mobility phenomena in the world (Facchini et al., 2019; Tombe and Zhu, 2019; Ma and Tang, 2020). Migrants leave their hometowns in pursuit of better employment opportunities, higher incomes, and improved living conditions, typically relocating to more economically developed regions. Within this context, we ask whether the impact of HSR on the quality of college admissions varies systematically with regional disparities in development. Addressing this question not only deepens our understanding of China's "Great Migration" but also offers potential insights into population, labor, and talent mobility in other countries. To examine this, we construct a measure of the economic development gap between the city where the college is located and the admission provinces. Using an interaction analysis approach, we find that the larger the bilateral economic development gap, the stronger the positive effect of HSR on college admission scores. We further exploit China's well-documented geographical imbalance in economic development—where eastern regions are significantly more developed than western regions. Leveraging the directional nature of the data, we find that the core effect identified in this paper is primarily driven by students from western regions applying to colleges in the east, as well as eastern students applying to colleges within the east. In contrast, the effect is smaller and statistically insignificant when students from the east apply to western colleges, or when students from the west apply to colleges within the west. These findings suggest that the HSR network may be exacerbating regional disparities in the quality of college admissions in China. While developed regions are able to leverage the HSR to attract higher-quality students, universities in developing regions derive relatively limited benefits.

In addition, because we are unable to obtain data on planned enrollment quotas or actual enrollment numbers of colleges (see Section 3.1 for a detailed discussion), we conduct supplementary empirical

[1] In this paper, "city" refers to China's four municipalities directly under the central government and prefecture-level cities. For convenience, we collectively denote them as cities.

[2] If a mechanism holds, introducing the relevant mechanism variable into the regression should block the causal pathway—manifested by a reduction in the magnitude and statistical significance of the core coefficient, potentially rendering it insignificant.

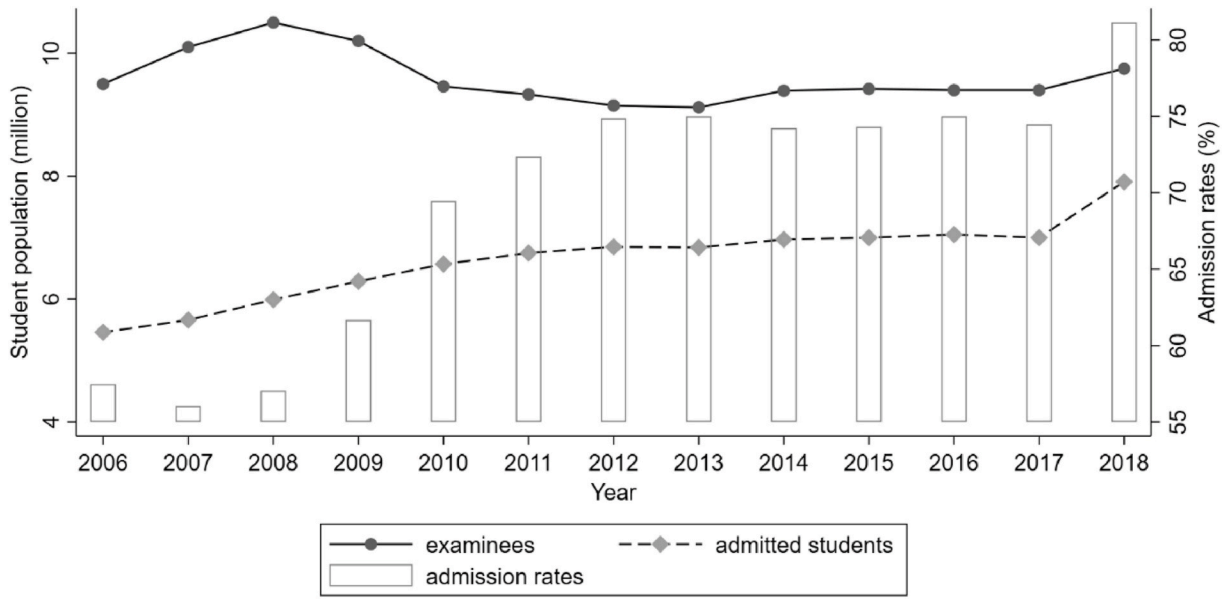

Fig. 1. Population of NCEE examinees and admission rates in China, 2006–2018.
Source: http://www.jdxzz.com/gkzx/2024/0806/10306262.html

analyses using the Chinese population census. The 2005 and 2015 Chinese population censuses report respondents' educational attainment status, which allows us to focus explicitly on the mobility of enrolled students. Drawing on the 2005 and 2015 population censuses and a gravity-style long-difference model, we find that HSR connectivity significantly increases the number of undergraduate and postgraduate students migrating from city *i* to city *j* for college attendance. Additionally, our analysis shows that when the western region becomes connected to the eastern region via HSR, the number of students from the western region attending colleges in the eastern region increase significantly. In contrast, when the eastern region is connected to the western region by HSR, there is no comparable increase in the magnitude or significance of flows. These empirical findings are consistent with those based on college admission scores, indicating that the HSR has widened regional disparities in China.

This paper contributes to several strands of the literature. The first is the extensive body of research on transportation infrastructure, particularly HSR. Existing studies have examined the impact of HSR from various perspectives, including, but not limited to, improvements in transportation accessibility (e.g., Cao et al., 2013; Shaw et al., 2014; Jiao et al., 2014; Chen and Haynes, 2017; Dong et al., 2020), the effects of HSR on firm behavior and performance (e.g., Xu, 2017; Charnoz et al., 2018; Bernard et al., 2019; Gao and Zheng, 2020), and the macroeconomic consequences of HSR (e.g., Chen and Haynes, 2015; Shao et al., 2017; Qin, 2017; Ke et al., 2017; Lin, 2017; Dong, 2018; Ahlfeldt and Feddersen, 2018; Yu et al., 2019). To the best of our knowledge, this paper is the first study to examine the impact of HSR on universities and colleges from the perspective of college admission scores. In doing so, we enrich the literature on transportation infrastructure by documenting a novel finding: HSR can enhance the quality of college admissions. In addition, we provide suggestive evidence that HSR may exacerbate regional disparities in college admissions quality. By approaching this issue from the perspective of college admissions quality, our study contributes new empirical evidence relevant to broader research on regional inequality and unbalanced development.

The second strand of literature is the growing body of research on NCEE. This literature can be broadly categorized into two main areas. The first leverages the unique institutional features of the NCEE to study student decision-making and admission mechanisms (e.g., Chen and Kesten, 2017; Chen and Kesten, 2019; Zivin et al., 2020; Wu and Zhong, 2020; Pu, 2021). The second area uses the NCEE as a contextual backdrop to examine topics such as higher education, labor market outcomes, gender disparities, rural-urban divides, and individual performance (e.g., Cai et al., 2019; Zivin et al., 2020; Li et al., 2024; Eble and Hu, 2025; Huang et al., 2025). For example, Zivin et al. (2020) find that higher temperatures on the day of the NCEE are associated with lower test scores, highlighting the role of environmental factors in cognitive performance. Li et al. (2024) provide evidence that the introduction of English listening tests in the NCEE widened the performance gap between urban and rural students, exacerbating educational inequality in China. Eble and Hu (2025) show that college name changes can lead to improvements in student quality, as reflected in higher admission scores. This paper contributes to the NCEE literature in two key ways. First, we provide empirical evidence that HSR enhances the quality of college admissions and widens regional disparities in admissions quality. Second, our findings suggest that external environmental factors surrounding universities and colleges—such as transportation infrastructure—play an important role in students' application decisions.

The third relevant strand of literature is the broader body of work on population and labor mobility, with a particular focus on talent mobility. In the context of this study, college students who pursue bachelor's degrees ultimately become part of the high-skilled labor force. Moreover, college graduates in China exhibit a strong tendency to remain in the region where they attended college. A survey of Chinese college students reports that 73.46% of graduates find employment in the same province where their college is located (Zhang and Zhou, 2018). Given the massive scale of China's higher education system and the NCEE, this implies that millions of students migrate away from their hometowns each year. The importance of talent is well recognized by the public, policymakers, and scholars alike—talent is a scarce resource, playing a central role in innovation and economic growth (Akcigit et al., 2024; Fan, 2024; Prato, 2025). Existing research has largely focused on the determinants of talent mobility, seeking to understand what factors attract skilled individuals (e.g., Solimano, 2008; Grogger and Hanson, 2015; Akcigit et al., 2016; Kerr et al., 2016; Gu et al., 2024). A key contribution of this paper lies in uncovering a new empirical channel for talent mobility: the expansion of the HSR network. By leveraging the institutional setting of the NCEE and admissions score data, as well as the Chinese population censuses, this paper provides novel empirical

evidence on how transportation infrastructure shapes patterns of talent mobility.

The remainder of the paper is organized as follows. Section 2 provides background information, introducing the institutional features of NCEE and the development of China's HSR network. Section 3 describes the data and empirical strategy, including the datasets used and the econometric models employed. Section 4 presents the main empirical results, including baseline regressions, parallel trends tests, placebo tests, and robustness checks. Section 5 explores potential mechanisms, examining three plausible channels: HSR's role in promoting city economic development, enhancing college strength, and improving transportation accessibility. Section 6 presents further analyses that explore the roles of bilateral distance and development gaps. It also provides supplementary empirical evidence based on the Chinese population census.

2. Background

Section 2.1 provides an overview of the institutional background of NCEE. Section 2.2 describes the development of China's HSR network since 2008. This background information lays the foundation for understanding the empirical strategy employed in this study.

2.1. NCEE in China

Although the NCEE is a national examination, all procedures are administered independently at the provincial level. The examination is administered in early June each year. Each province may choose to design its own test papers or adopt those provided by the Ministry of Education. During our study period (2006 to 2018), the NCEE consisted of two tracks: humanities and science.[3] Each candidate could only choose one track to take the exam, and all subsequent grading and admissions processes were conducted separately for each track. After the exam, each province organizes a closed, standardized grading process. Every student's answer sheets are scored under a unified and fixed rubric. Graders are typically experienced teachers with extensive classroom backgrounds, and they are required to follow strict scoring guidelines. Additional auditing procedures are in place to ensure compliance.[4] This rigorous institutional setup ensures that within the same year, province, and track, NCEE scores are highly comparable across students.

Candidates consider their scores and personal preferences to select target colleges through an official application system. This selection process is known as the college preference submission. The application process is divided into three admission rounds: Batch 1, Batch 2, and Batch 3 of undergraduate admissions. Students are free to apply in any one of these batches, and colleges may participate in recruitment across different batches. The official provincial application system is responsible for matching applicants with admissions. For candidates applying to the same college within the same batch, admissions are granted in descending order of scores until the college's quota is filled or all applicants are admitted. Finally, colleges issue admission letters to successful applicants, who then enroll at the end of August or early September.

It is important to note that during the study period in this paper, the NCEE admission process underwent major institutional reforms, two of which are particularly salient. First, the "post-score preference submission" reform shifted the timing of preference submission from before candidates learned their NCEE scores to after scores were released (Ha et al., 2020). When candidates were required to submit preferences without knowing their NCEE scores, they could only guess or estimate their NCEE scores, which introduced substantial uncertainty into the application process. To reduce application risk faced by candidates, provinces gradually implemented the "post-score preference submission" reform. Appendix Table A1 reports the timing of this reform across Chinese provinces. Second, the "parallel preference" reform replaced the Boston mechanism with a parallel mechanism in the admission process (Chen and Kesten, 2017). Although both mechanisms allow candidates to submit multiple preferences, they differ substantially. Under the Boston mechanism, submitted preferences are strictly ordered. Because colleges prioritize applicants who list them as their first choice, second and lower-ranked preferences are largely ineffective. As a result, candidates face a dilemma: setting an overly ambitious first choice risks admission failure, while choosing a conservative first choice may lead to underutilization of a high NCEE score. In contrast, under the parallel mechanism, all submitted preferences are treated as having equal priority. Candidates can list more selective colleges as reach options while including less selective institutions as safety choices. Appendix Table A2 reports the timing of the "parallel preference" reform across Chinese provinces.

In Appendix Figure A1, we present the time trend of the average number of provinces from which colleges recruit. We find that the average number of recruiting provinces remains stable at approximately 22–23 provinces during the study in this paper. Appendix Figure A2 illustrates the spatial variation in the average number of recruiting provinces. From a spatial perspective, this measure is relatively evenly distributed. These patterns suggest that although the NCEE has undergone several major reforms, such reforms are typically implemented at the provincial level. Consequently, from the perspectives of both time trend and spatial variation in the average number of recruiting provinces, these reforms do not appear to have a pronounced impact on colleges' recruitment scope. In Section 4.3, we further demonstrate empirically that our core results remain robust after controlling for the "post-score preference submission" reform and the "parallel preference" reform.

Except for a very small number of special cases, high school graduates can only obtain college admission qualification through the NCEE, which grants access to either regular undergraduate programs or regular vocational programs. Regular undergraduate students receive a general education within disciplines and, upon meeting graduation requirements, are awarded both a diploma and a bachelor's degree. Regular vocational students receive specialized skill training in specific fields and are awarded a diploma upon completion but do not receive a bachelor's degree. On average, regular undergraduate students achieve higher NCEE scores than vocational students. In addition to entering the labor market, regular undergraduate graduates are eligible to take postgraduate entrance exams for master's and doctoral programs, whereas vocational graduates typically enter the workforce directly. In this paper, we focus on regular undergraduate programs. Fig. 2 presents the trends in enrollment, current student, and graduations for regular undergraduate programs in China during the study period (2006 to 2018). The figure illustrates the large and growing scale of the regular undergraduate student body in China, with the numbers of graduates, enrolled students, and admissions approximately doubling from 2006 to 2018. Appendix Figure A3 shows corresponding trends for regular vocational programs over the same period, and Appendix Figure A4 compares overall changes in both regular undergraduate and vocational programs. The qualitative patterns are broadly similar.

[3] Both the humanities and science require students to take Chinese, Mathematics, and a foreign language. The primary distinction between the two subjects lies in the specialized subjects: students in the humanities subject study history, politics, and geography, while those in the science subject study physics, chemistry, and biology. In addition, the mathematics exam differs between the two subjects—not only are the test papers distinct, but the level of difficulty is also lower in the humanities compared to the science.

[4] For example, the essay section of the NCEE Chinese exam is scored independently by two graders. If the score difference between them exceeds a predetermined threshold, the essay is then evaluated by a third independent grader. Should the discrepancy remain beyond the threshold, the essay is subsequently reviewed and assessed by a specialized committee to determine the final score.

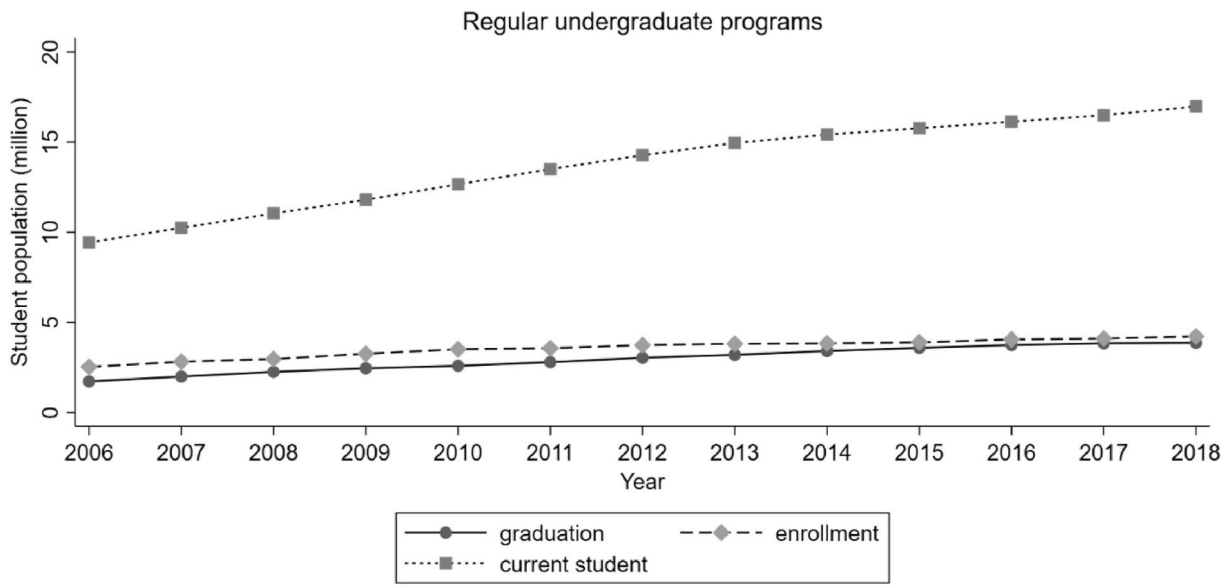

Fig. 2. Student population of regular undergraduate programs in China, 2006–2018.
Source: *China Education Statistical Yearbook*

2.2. HSR in China

Since 2008, the development of China's HSR network has been remarkably rapid. Appendix Figure A5 illustrates the evolution of total HSR operational mileage during the study period (2006 to 2018). In 2008, China's HSR operational mileage was 671.5 km, increasing to 29,903.8 km by 2018—an average annual growth rate of 46.17%. The share of HSR in total railway operational mileage rose from 0.8427% to 22.7144% over the same period. Appendix Figures A6 and A7 show the changes in HSR passenger volume and passenger turnover, respectively. The average annual growth rate of HSR passenger volume was approximately 75.67%, with its share in total railway passenger volume increasing from 0.5% to 60.8693%. For passenger turnover, the average annual growth rate was around 83.82%, and HSR's share increased from 0.2% to 48.5765%.

Appendix Figures A8 and A9 present the city-level distribution of HSR opening in China in 2008 and 2020, respectively. Appendix Figure A8 shows that in 2008, only a handful of cities had access to HSR. In contrast, Appendix Figure A9 illustrates that by 2018, the majority of cities in China's central and eastern regions had been integrated into the HSR network. The rapid expansion of China's HSR system over time and across space provides substantial variation for empirical identification in this study.

3. Data and empirical design

3.1. Data

We obtained data on college admission scores from 2006 to 2018 from Sina Gaokao. This dataset includes average and maximum admission scores at the level of college-year-admission province-track (humanities or science)-admission batch (Batch 1, Batch 2, and Batch 3 undergraduate programs). Following the approach of Eble and Hu (2025), we exclude colleges with military affiliations from our analysis. In addition, we remove colleges associated with Hong Kong and Macau. The processing of the college admission score data proceeds as follows. First, we exclude observations with average scores exceeding those of Peking University or Tsinghua University.[5] Second, we retain only those observations with average scores ranging between 200 and 750.[6] These final two steps are primarily intended to eliminate outliers. It should be noted that we do not have access to planned enrollment quotas or actual enrollment numbers data of colleges.[7] In Section 4.3, we discuss the robustness of our results to potential confounding from changes in planned enrollment. Additionally, in Section 6.3, we use Chinese population census data to analyze the impact of HSR on the mobility of highly educated individuals and enrolled students, thereby complementing the analysis based on college admission score data. For further details and an introduction to the Chinese population census data, see Sections 4.3 and 6.3.

We use the *List of National Regular University and College (2020)* published by the Ministry of Education of China (hereafter referred to as the "College List") as the primary reference to identify a unique identifier for each college, supplementing the list with manual data collection from online sources. Data on college-level information are obtained

[5] This is because Peking University and Tsinghua University are the most prestigious and elite universities in China. In theory, they consistently have the highest admission thresholds in the country. Each year, these two universities compete intensely to attract the top NCEE scorers—those who rank first in each subject and province—to enroll in their programs. This phenomenon is commonly referred to as the "Battle for the Top Scorer" (Xie and Bi, 2015).

[6] During the study period of this paper, the total score of the NCEE is 750 points. It is highly uncommon for the admission scores for regular undergraduate programs to fall below 300 points. Scores lower than this threshold typically correspond to vocational programs. Therefore, within the context of this study, which focuses on regular undergraduate institutions, a lower bound of 200 points is a deliberately conservative threshold.

[7] In our attempts to obtain college admission scores, we did not find that Sina Gaokao provides data on planned enrollment quotas or actual enrollment numbers data of colleges. To our knowledge, another potential source is China Education Online (https://www.eol.cn/). However, China Education Online only provides admission scores and planned enrollment quotas from 2020 onward. For the context of this study, China's large-scale HSR expansion occurred over the decade following 2008 (Fang et al., 2025). After 2020, HSR construction may lack sufficient variation for identification.

from the *Compilation of Science and Technology Statistics of University and College* (2007–2017), also published by the Ministry of Education (hereafter referred to as the "College Compilation"). The *College Compilation* provides institution-level information on personnel, funding, research projects, and scientific outputs. In this study, we merge the three datasets with the college as the core unit of analysis.

HSR data are obtained from the CNRDS database, which includes information on the opening dates of HSR stations, their respective cities, and associated HSR lines. Based on this database, we construct a city-level binary variable indicating whether a city is connected to the HSR network. City-level meteorological data are sourced from the China Meteorological Data Service Centre. City statistical data are drawn from the *China City Statistical Yearbook*. For missing values in certain variables, we impute data using the average of adjacent periods. All relevant variables are deflated using the year 2000 as the base year.

As noted in Section 2.1, NCEE scores are comparable within the same year-admission province-track group, but not across different combinations of year, admission province, and track. To enable comparisons of college admission scores across different year-admission province-track group, we follow the approach of Li et al. (2024) and Eble and Hu (2025) by nondimensionalizing the NCEE scores at the year-admission province-track level, as shown in Equation (1).

$$average_n_{ipsbt} = 100 \times \frac{\left[average_{ipsbt} - mean_{pst}(\cdot)\right]}{\left[\max_{pst}(\cdot) - \min_{pst}(\cdot)\right]} \tag{1}$$

Let i denote the college, p the admission province, s the track, b the admission batch, and t the year. The variable $average_{ipsbt}$ represents the average admission score for college i in year t, admission province p, track s, and batch b. The terms $mean_{pst}(\cdot)$, $\max_{pst}(\cdot)$, and $\min_{pst}(\cdot)$ refer to the mean, maximum, and minimum values of $average_{ipsbt}$ within the corresponding year-admission province-track group. The normalized score, denoted as $average_n_{ipsbt}$, is the result of normalizing $average_{ipsbt}$ to the range [0,100] within each year-admission province-track group. We use this normalization procedure to rank college admission scores in ascending order. Multiplying the normalized values by 100 allows us to interpret the normalizing score as a percentile rank of a college's admission threshold within the same year-admission province-track group. This facilitates interpreting the regression coefficients as percentage point changes in a college's percentile ranking. In the robustness checks, we demonstrate that our results remain consistent when alternative nondimensionalization methods (e.g., standardization), different normalizing levels (e.g., year-admission province-track-batch), or the raw college admission scores without nondimensionalization are used.

Considering that the NCEE is administered mid-year, we apply the following procedure when constructing the key independent variable — the binary variable (*hsr*) of whether the HSR is opening or not at the city level. If a city's HSR opens in the first half of the year (January to June), the variable *hsr* is coded as 1 starting from that same year; otherwise, it is coded as 0. Conversely, if the HSR opens in the second half of the year (July to December), the variable *hsr* takes the value 1 beginning the following year; otherwise, it remains 0. In robustness checks, we confirm that the results hold robustness when using the previous year's HSR status for the city.

3.2. Empirical design

In order to explore the impact of HSR opening on the college admissions quality, we set up a DID model based on existing literature (e.g., Andersson et al., 2023; Berger and Prawitz, 2024) as shown in Equation (2).

$$average_n_{cipsbt} = \alpha + \beta hsr_{c,t} + \boldsymbol{\gamma}\mathbf{X}_{c,t} + \delta_{ipsb} + \mu_{psbt} + \varepsilon_{cipsbt} \tag{2}$$

Here, c denotes the city in which college i is located. $average_n_{cipsbt}$ represents the normalized average admission score for college i. $hsr_{c,t}$ is a binary variable indicating whether HSR is opening in city c in year t. The vector $\mathbf{X}_{c,t}$ includes a set of control variables. We incorporate college-admission province-track-batch fixed effects δ_{ipsb} and year-admission province-track-batch fixed effects μ_{psbt} in the regression. ε_{cipsbt} is the random error term. Robust standard errors are clustered at the city level.

We include the following control variables in our analysis: city economic scale, measured by the logarithm of GDP; city population density, calculated as total population divided by administrative land area; city industrial structure, captured by the shares of secondary and tertiary industry value-added; and city higher education capacity, proxied by the total number of colleges and the logarithms of the total number of colleges teachers. These control variables help mitigate potential omitted variable bias and improve the precision of the estimates. For instance, both city economic scale and population density may simultaneously influence the college admission scores and the likelihood of HSR opening. The college-admission province-track-batch fixed effects δ_{ipsb} control for all time-invariant characteristics at the college-admission province-track-batch level, such as bilateral geographic distance, historical and cultural ties, and climate differences. The year-admission province-track-batch fixed effects μ_{psbt} account for all time-varying characteristics at the admission province-track-batch level, such as changes in NCEE-related policies implemented by admission provinces across years, subjects, and batches. In the robustness checks, we incorporate additional control variables and adopt more stringent fixed effects specifications, and the results remain robust.

In the context of this study, the treatment group consists of cities with HSR opening, while the control group includes cities without HSR. The difference in college admission scores between the treatment and control groups, before and after HSR opening, captures the causal effect of interest—represented by the estimated coefficient $\widehat{\beta}$. This coefficient reflects the percentage-point change in the percentile ranking of a college's admission score attributable to the opening of HSR in the city. To provide preliminary evidence on the impact of HSR on college average admission scores, we divide the sample into two groups based on HSR opening and plot the density distributions of college average admission scores. As shown in Fig. 3, on average, colleges located in cities with HSR tend to have higher average admission scores than those in cities without HSR. In other words, the opening of HSR is associated with an upward shift in average admission scores. Descriptive statistics for the main variables are reported in Appendix Table B1.

4. Empirical results

4.1. Baseline regression

Table 1 reports the regression results corresponding to Equation (2). In column (1), we control for two-way fixed effects, namely college-admission province-track-batch fixed effects and year fixed effects, without including any control variables. In column (2), we add the full set of control variables. Columns (3) and (4) replace the year fixed effects with more stringent year-admission province-track-batch fixed effects. Across all four columns in Table 1, the coefficient on the key independent variable *hsr* is consistently positive and statistically significant. This indicates that the opening of HSR in a city significantly increases the quality of college admissions, as reflected in higher average admission scores of local colleges. Specifically, the estimate in column (4) suggests that HSR access raises the average admission score by 0.372 percentage points. We adopt the specification in column (4) as our baseline model for subsequent analyses.

The existing literature has highlighted that traditional two-way fixed effects (TWFE) DID models can yield severely biased estimates in the presence of heterogeneous treatment effects (e.g., De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). To address this issue, we adopt the estimation strategies proposed by Sun and Abraham (2021) and Cengiz et al. (2019). Panel B and Panel

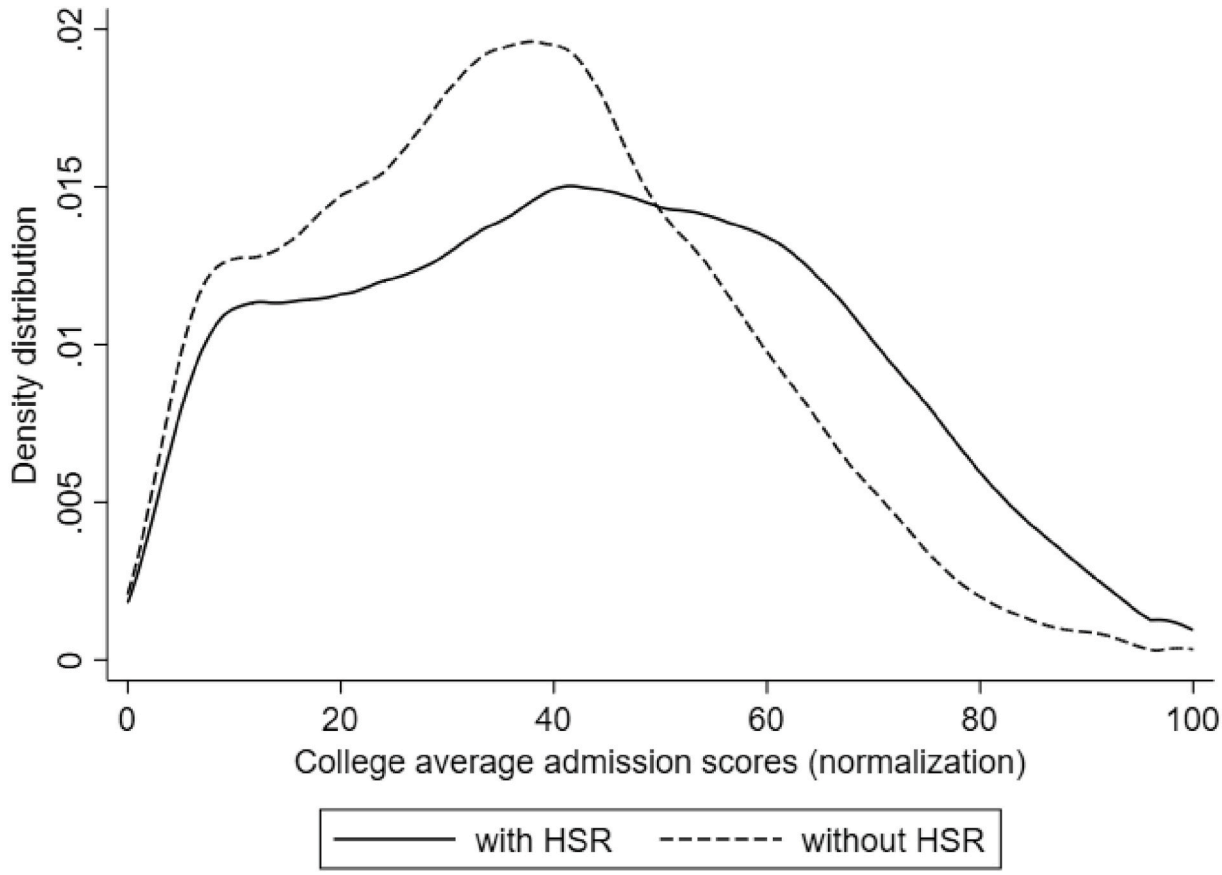

Fig. 3. Density distributions of college average admission scores.

C of Table 1 report treatment effects estimated using the methods proposed by Sun and Abraham (2021) and Cengiz et al. (2019), respectively. The results in Panels B and C show that, under both estimation approaches, the opening of HSR in a city significantly increases colleges' average admission score.

We now turn to a discussion of the economic significance of the core effect estimated in this paper. Given the unique characteristics of NCEE, we argue that an increase of 0.372 percentage points is economically meaningful. As discussed in Section 2.1, college admissions under the NCEE are determined strictly by ranking students according to their NCEE scores in descending order. Due to the highly competitive nature of this matching mechanism, even a slight increase in scores allows students to gain admission to more prestigious colleges (Li et al., 2024). Conversely, even a marginal improvement in percentile rank enables colleges to enroll higher-quality students. Competition among Chinese colleges is intense, with institutions striving to attract top-performing candidates (Chen et al., 2022). For example, colleges actively engage in extensive promotional campaigns across all provinces, offering generous scholarships and flexible academic programs to attract high-aptitude students (Xie and Bi, 2015; Wang, 2024).

Moreover, given the large number of colleges and the vast scale of the NCEE candidate population in China, the estimated effect is likely to be even more economically significant. Among the general public in China, there is a well-known slogan regarding the NCEE: "Gaining one more point defeats a thousand others." This phrase implies that a one-point increase in NCEE scores can result in surpassing one thousand candidates in the ranking.[8] We can conduct a rough calculation to assess the plausibility of this claim. Assuming that approximately 10 million students take the NCEE each year across 34 provinces, with two track per province, this implies roughly 150,000 candidates competing in each track within a province. If we assume these scores are normally distributed over the [200, 750] range, then around the middle of the distribution, an increase of just one point could realistically correspond to a jump of about 1000 ranks. To provide empirical support for this claim, we use 2018 data from the Anhui province's science track, sourced from the Gaokao website (https://www.gaokao.cn/), which reports the number of candidates achieving each score. Appendix Figure C1 plots the number of candidates at each score level. It shows that between scores of 440 and 480, more than 1000 students share each individual score, and between 360 and 580, each score point corresponds to more than 500 students. In addition, college rankings in China are relatively rigid and hierarchical.[9] Each college typically competes only with peer institutions of similar strength. In our sample, there are 1215 undergraduate universities.[10] A 0.372 percentile increase corresponds to an improvement of approximately 4 to 5 positions in the national ranking. In the highly rigidly stratified Chinese higher education system, such an upward shift in rank is far from negligible.

Finally, it is useful to compare the findings of this paper with those in the existing literature. Li et al. (2024) provide empirical evidence that the introduction of an English listening component to the NCEE resulted in a 1.1 percentile point decline in rural students' total NCEE scores,

[8] https://cj.sina.com.cn/articles/view/6321657948/178ccd85c02001fdlk.

[9] In China, there exists a widely recognized and rigidly hierarchical university and college ranking system. Peking University and Tsinghua University occupy the top tier. They are followed, in descending order, by universities classified under the "985 Project," the "211 Project," first-batch universities, second-batch universities, and third-batch universities.

[10] This number aligns with the number observed in prior studies, such as Eble and Hu (2025), who analyze a set of 1198 colleges.

Table 1
Baseline regression results.

	(1)	(2)	(3)	(4)
Panel A	Baseline (TWFE)			
hsr	0.788***	0.557***	0.436***	0.372***
	(0.172)	(0.163)	(0.117)	(0.110)
N	424,867	416,173	424,822	416,124
R-squared	0.893	0.895	0.968	0.969
Panel B	Sun and Abraham (2021)			
hsr	1.189***	0.816***	0.577***	0.410***
	(0.192)	(0.212)	(0.115)	(0.119)
N	424,867	416,173	424,822	416,124
R-squared	0.8929	0.8949	0.9682	0.9687
Panel C	Cengiz et al. (2019)			
hsr	2.716***	2.117***	1.491***	1.086***
	(0.336)	(0.348)	(0.316)	(0.326)
N	1,437,271	1,373,233	1,436,673	1,372,651
R-squared	0.892	0.896	0.971	0.972
Controls		✓		✓
College-admission province-track-batch (-stack) FE	✓	✓	✓	✓
Year-admission province-track-batch (-stack) FE			✓	✓
Year (-stack) FE	✓	✓		

Note: The dependent variable is the average admission score at the college-admission province-track-batch-year level. The key independent variable *hsr* is a city-level binary indicator denoting whether HSR is opening. We implement the heterogeneous treatment effect estimation method proposed by Sun and Abraham (2021) using the Stata command *eventstudyinteract*. For the stacked DID approach of Cengiz et al. (2019), we define the never-treated units—cities that did not open HSR at any point during the study period—as the pure control group. For each cohort of treatment events, we construct a new subsample consisting of the treated group and the pure control group. All such subsamples are then stacked to form the final stacked sample. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

accounting for nearly 30% of the baseline urban–rural gap in college admissions in China. The core effect identified in this paper is approximately one-third the magnitude of that found by Li et al. (2024). Eble and Hu (2025) also utilize college NCEE admission score data. Their empirical analysis shows that college name changes led to an increase in average admission scores of 0.057 to 0.077 standard deviations. It is worth noting that we employ a more stringent set of fixed effects in this paper than Eble and Hu (2025). When we adopt the same fixed effects as Eble and Hu (2025)—including college, admission province, year, and track fixed effects—the estimated effect of HSR opening corresponds to a 0.188 standard deviation increase in average admission scores. Under this specification, the core effect identified in this paper is roughly one-third to one-quarter the size of the effect documented by Eble and Hu (2025). Thus, the economic significance of our findings is comparable to that of existing studies in the literature.

4.2. Parallel trend test and placebo test

Identifying causal effects based on the DID approach requires the parallel trends assumption to hold. Specifically, conditional on the set of control variables and fixed effects, the treatment and control groups should exhibit no significant difference in outcome trends prior to the shock. To provide empirical evidence supporting the parallel trends assumption in this study, we employ the commonly used event-study methodology in the literature, as specified in Equation (3).

$$average_n_{cipsbt} = \alpha + \sum_{k=-5}^{5} \beta_k impact_{c,t=k} + \boldsymbol{\gamma}\mathbf{X}_{c,t} + \delta_{ipsb} + \mu_{psbt} + \varepsilon_{cipsbt} \tag{3}$$

The term $\sum_{k=-5}^{5} impact_{i,t=k}$ represents a set of event-time dummy variables indicating the number of periods t relative to the timing of HSR opening k. For example, $k = -5$ refers to five periods before the opening of HSR, while $k = 0$ denotes the current year of HSR opening. All other specifications—including control variables, fixed effects, and the clustering level of standard errors—remain the same as in Equation (2). We normalize the period immediately prior to the HSR opening, $k = -1$, as the reference category. Fig. 4 reports the estimated coefficients β_k and 95% confidence intervals for Equation (2). To address the heterogeneous treatment effects issue, we adopt the estimation strategies proposed by Sun and Abraham (2021) and Cengiz et al. (2019). Fig. 4 presents the corresponding event-study results. The results presented in Fig. 4 show no evidence of significant differences in trends between the treatment and control groups prior to the opening of HSR. This supports the validity of the parallel trends assumption underlying our DID approach. In addition, the estimates after HSR opening indicate that the core effect increases gradually over time and remains consistently positive and statistically significant. This suggests that the impact of HSR on college admissions quality is persistent and grows over time.

Finally, one may be concerned that the observed positive effect of HSR on college average admission scores is merely coincidental. For instance, it is possible that certain unobserved shocks happened to raise admission scores in cities where HSR was opening. To address this concern, we conduct a placebo test. Specifically, we randomly assign HSR treatment status at the city level and re-estimate the regression model, repeating this procedure 500 times. We expect that the vast majority of placebo estimates should be smaller than the baseline coefficient. Appendix Figure C2 plots the distribution of the estimated coefficients from the 500 placebo regressions. The results show that the placebo coefficients are approximately normally distributed and centered around zero. Importantly, all placebo estimates are smaller than the baseline estimate of 0.372 reported in Column (4) of Table 1, as indicated by the vertical line in Appendix Figure C2. This suggests that the estimated positive effect of HSR opening on college admissions quality is unlikely to be driven by coincidence.

4.3. Robustness checks

In the baseline regression, we follow Li et al. (2024) and Eble and Hu (2025) by normalizing average admission scores at the year-admission province-track level. This approach facilitates comparison of scores across year-admission province-track groups. To assess the robustness of our core findings, we adopt alternative data processing strategies. Appendix Table C1 presents the regression results. In Column (1), instead of normalization, we standardize the dependent variable to non-dimensionalize. Specifically, the dependent variable is the standardized average admission score at the year-admission province-track level. The results in Column (1) indicate that the opening of HSR in a city is associated with a 0.0121 standard deviation increase in the average admission score. In Column (2), we redefine the dependent variable as the normalized average admission score at the year-admission province-track-batch level. In Column (3), we use the raw average admission score as the dependent variable. The results show that following the opening of HSR, the average admission score increased by 1 point. In the final two columns, we examine the effect of HSR on the maximum admission scores. In Column (4), the dependent variable is the normalized maximum admission score at the year-admission province-track level. In Column (5), we use the raw maximum admission score. Results from both columns indicate that the opening of HSR significantly raises the maximum admission score. This suggests that HSR not only attracts average students but also draws in highest-ability students—a positive outcome for Chinese colleges facing intense

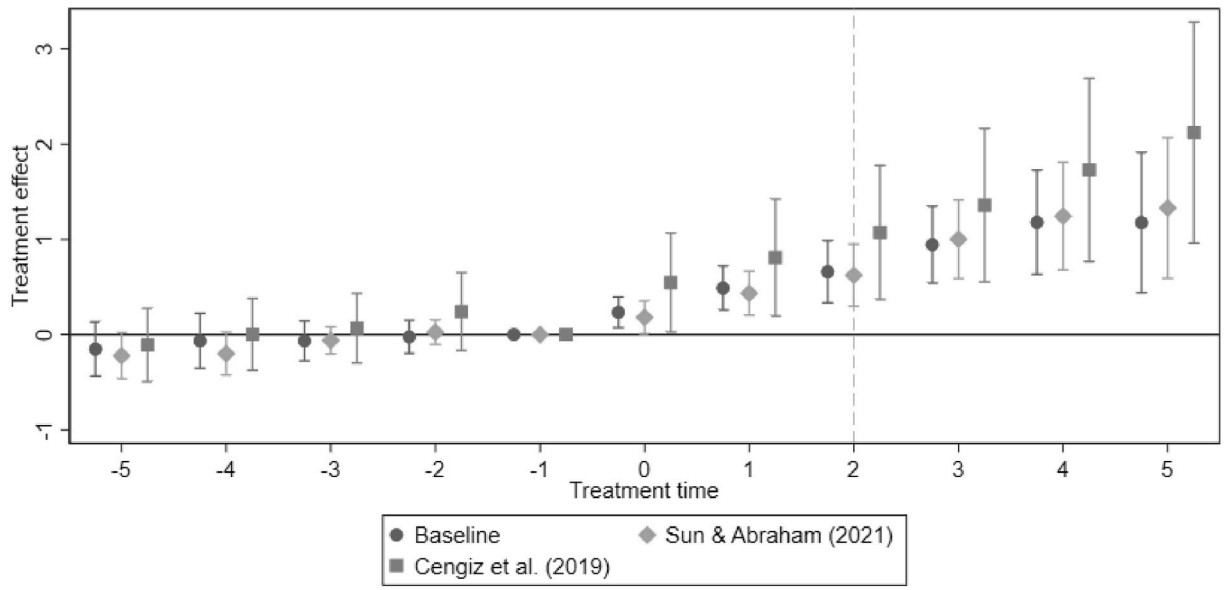

Fig. 4. Event Study Results
Note: Fig. 4 shows the estimated coefficients β_k and 95% confidence intervals for Equation (3) and the event study results using the methods provided by Sun and Abraham (2021) and Cengiz et al. (2019).

competition.

Appendix Table C2 presents the second set of robustness checks. In Column (1), we retain all observations, including those where the average admission score exceeds 750 or falls below 200. In Column (2), building upon Column (1), we further retain observations with average admission scores higher than those of Peking University or Tsinghua University. Columns (3) and (4) apply more stringent data restrictions. In Column (3), we narrow the range of average admission scores to [300, 700]. In Column (4), we winsorize the average admission scores at the 1st and 99th percentiles. To account for possible college name changes over time, we construct unique college identifiers based on *College List*. However, a potential concern is that examinees may perceive pre- and post-renaming colleges as distinct. To address this, in Column (5) of Appendix Table C2, we generate college identifiers directly from college names as they appear in the raw data. The results reported in Appendix Table C2 confirm that our findings remain robust across all specifications.

Appendix Table C3 reports the third set of robustness checks. In the baseline regressions, both the key independent variable and control variables are measured contemporaneously. However, one may be concerned that candidates primarily consider city characteristics from the previous year when making decisions. To address this, in Column (1), we replace the contemporaneous control variables with their one-year lagged counterparts. In Column (2), we include both contemporaneous and lagged control variables. Given that the NCEE takes place in the middle of each year, our baseline specification defines the key independent variable based on whether HSR service commenced in the first half of the year. In Column (3), we instead define HSR openings at the annual level and lag the variable by one year to simulate the possibility that candidates respond to whether a city had HSR openings in the previous year. In Column (4), we apply a more conservative approach to inference by clustering robust standard errors at the college's provincial level. To account for the fact that the construction of China's HSR network began in 2008, Column (5) restricts the sample period to 2008–2018. In Column (6), we exclude Beijing and Shanghai from the sample, as these two major metropolitan areas are economically advanced, serve as national transportation hubs, and host a disproportionately large number of colleges. Across all specifications, the results presented in Appendix Table C3 continue to support the robustness of our core findings.

Appendix Table C4 incorporates additional control variables and more stringent fixed effects specifications. In Column (1), we control for the logarithm of city-level railway passenger volume, civil aviation passenger volume, and road passenger volume to account for the potential confounding effects of other modes of transportation, including ordinary rail, air travel, and road transport. In Column (2), we further include city-level annual meteorological variables, such as average wind speed, precipitation, temperature, atmospheric pressure, sunshine duration, and relative humidity. In Column (3), we add city-specific linear time trends to control for all city-level time-varying linear characteristics. In Column (4), we replace the baseline year-admission province-track-batch fixed effects with a more stringent set of fixed effects: college province-year-admission province-track-batch, to absorb all time-varying factors at the level of college province-admission province-track-batch. Finally, in Column (5), we simultaneously include all additional controls and fixed effects used in Columns (1) through (4). The results presented in Appendix Table C4 demonstrate that our main findings remain robust under these more demanding specifications.

In Section 2.1, we introduced two major NCEE reforms: the "post-score preference submission" reform and the "parallel preference" reform. To account for the potential influence of these reforms, we restrict the sample to provinces whose admissions policies remained unchanged during the study period (2006–2018). Appendix C provides a detailed explanation. Columns (1) and (2) of Appendix Table C5 report the corresponding regression results. The results indicate that our core findings remain robust after controlling for the effects of these two reforms. We acknowledge that it is not feasible to discuss all NCEE reforms. It is important to note that the NCEE procedures are implemented independently at the provincial level, which implies that most reforms are also province-specific. In our econometric model, we include year–admission province–track–batch fixed effects. These fixed effects absorb all time-varying changes at the admission province–track–batch level, including all admission province-specific reforms. Consequently, the potential interference of these NCEE reforms with our core estimates is likely minimal.

As discussed in Section 3.1, we do not have access to planned enrollment data of colleges during the study period (2006–2018). Therefore, we cannot fully rule out potential confounding from changes in colleges' enrollment plans, such as entering or exiting specific

recruiting provinces or variations in enrollment quotas. Cinelli et al. (2024) and Whited et al. (2022) note that control variables that only affect the dependent variable are neutral variables: including them in regressions improves estimation precision but does not affect the causal pathway. The planned enrollment at the college–admission province level may influence admission scores but is unlikely to affect whether a city opens HSR. Thus, planned enrollment affects only the dependent variable, not the core independent variable. In theory, conditional on the significance of our core effect, controlling for planned enrollment would not materially alter the causal identification in this study. We acknowledge that we cannot fully rule out sample selection issues caused by colleges entering or exiting specific admission provinces due to data restriction. This awaits further research supported by breakthroughs in data availability.

Finally, concerns may arise that differences in unit size could affect the estimation results. One common approach to enhance robustness is to weight the regressions by unit size (e.g., enrollment quotas or number of admitted students). Although we do not have access to enrollment quotas and admitted students for weighting, the *China City Statistical Yearbook* provides city-level data on college student populations and teachers numbers. We use the number of college students and the number of college teachers in each city as regression weights, thereby giving greater explanatory importance to cities with larger higher education systems. Columns (3) and (4) of Appendix Table C5 report the corresponding regression results. We find that the results remain robust when weighting by city-level college student and teacher numbers. In Columns (5) and (6) of Appendix Table C5, we further conduct weighted regressions using the lagged values of city-level college student and teacher numbers, and the results remain robust.

5. Potential mechanisms analysis

In this section, we explore the potential mechanisms through which HSR improves the quality of college admissions. The observed increase in admission scores reflects a shift toward candidates with stronger performance on the NCEE. A natural question arises: what factors are attracting these higher-performing candidates? We propose three plausible mechanisms in the context of this study. First, HSR may promote local economic development, and candidates may prefer cities with higher levels of economic prosperity. Second, HSR may enhance the overall strength of college, which in turn attracts more capable students. Third, candidates may be drawn by the improved transportation accessibility brought by the HSR. We present evidence suggesting that the first two mechanisms cannot account for our main findings, whereas the third mechanism can. In this section, we briefly present the empirical test results for the first two mechanisms, while the detailed analysis is provided in Appendix D. The primary focus of this section is on the third mechanism.

For the first mechanism, we construct a set of city-level indicators that may influence candidates' preferences, covering income, consumption, international economic activity, and public services. For the second mechanism, we construct a set of college-level strength indicators that may attract candidates, including measures of personnel, funding, and research output. We employ two approaches for the empirical tests. First, following the approach of Chor et al. (2021) and Dippel et al. (2022), we include the potential mechanism variables as additional control variables in the baseline regression. Second, we empirically examine the impact of HSR opening on these potential mechanism variables. The results reported in Appendix Figures D1–D6 show no evidence that the first two mechanisms can explain our core findings. A detailed discussion of the empirical tests for these two mechanisms is provided in Appendix D. We now turn to the analysis of the third mechanism.

A third potential mechanism through which HSR improves the quality of college admissions is the enhanced transportation convenience it brings. HSR offers significant advantages in terms of speed, travel time, and long-distance connectivity. Existing literature suggests that HSR increases regional accessibility and reduces travel time. For instance, Cao et al. (2013) measured weighted average travel times (WATT) and potential accessibility (PA) across mainland China using multiple indicators. Their findings show that, compared to conventional railways, most cities experienced improvements exceeding 70% in both WATT and PA due to HSR. Shaw et al. (2014) reached similar conclusions. For major HSR routes, travel time was reduced from 10 to 12 h via conventional railways to 3–5 h with HSR. Shaw et al. (2014) also found that, between 2008 and 2013, HSR led to an overall 15.10% reduction in business travel costs across mainland China. Liu and Zhang (2018) further demonstrate that HSR enhanced accessibility for all Chinese cities, reducing the national average inter-city travel time by 45%. Overall, the development of the HSR network substantially improves urban accessibility and strengthens interregional connectivity (Jiao et al., 2014; Chen and Haynes, 2017).

First, we empirically examine the heterogeneous effects associated with cities' positions within the HSR network. The corresponding regression results are reported in Appendix Table DX. We find that the significant positive effect of HSR on admission scores is primarily driven by three types of cities. First, cities where HSR stations are located in the urban districts. Second, cities served by multiple HSR lines. Third, cities that serve as key nodes in the HSR network—either as departure or terminal. One potential concern is that the number of HSR lines passing through a city and whether a city is located at the origin or terminus of an HSR line may simply reflect the city's economic strength and political status. Additionally, we acknowledge that the above three heterogeneity analyses do not fully capture the transportation convenience benefits brought by HSR. We are more inclined to treat these heterogeneity results as suggestive evidence. Therefore, the detailed analysis is provided in Appendix D, and only a brief overview is presented here. More importantly, we adopt an alternative mechanism analysis approach. Specifically, we construct a bilateral HSR connectivity measure between the college city and the admission province. Based on a gravity-type econometric model, our empirical analysis finds that bilateral HSR connectivity significantly increases college admission scores.

Specifically, we use HSR line information from CNRDS to construct the HSR network in China. We define a bilateral HSR connectivity dummy variable: it takes a value of 1 if two regions are connected within the same HSR network (i.e., one region can reach the other via one or more uninterrupted HSR lines), and 0 otherwise. Based on this, we construct the gravity-type econometric model in Equation (4).

$$average_n_{cipsbt} = \alpha + \beta connect_{c,p,t} + \delta_{ipsb} + \mu_{isbt} + \tau_{psbt} + \varepsilon_{cipsbt} \tag{4}$$

Here, variable $connect_{c,p,t}$ denotes the bilateral HSR connectivity between college city c and admission province p. δ_{ipsb} represents college–admission province–track–batch fixed effects, μ_{psbt} represents college–track–batch–year fixed effects, and τ_{psbt} represents admission province–track–batch–year fixed effects. The fixed effects μ_{psbt} and τ_{psbt} absorb all time-varying unilateral characteristics, controlling for the push–pull factors in the gravity model. The fixed effects δ_{ipsb} absorb all time-invariant bilateral characteristics, such as geographic distance and cultural differences. ε_{cipsbt} is the random error term. Robust standard errors are clustered at the college city–admission province level.

We first define bilateral HSR connectivity between college city and admission province as follows: if the college city and any city in the admission province are located in the same HSR network, there is an HSR connection between the college city and the admission province. Column (1) of Table 2 reports the regression results. We find that HSR connectivity between a college city and an admission province has a positive effect on the college's admission scores in that province; however, the estimated coefficient is statistically insignificant and imprecise. One possible explanation is that two regions in the same HSR network but on different HSR lines may not experience substantial improvements in transportation convenience.

Table 2
Regression results based on the bilateral HSR connectivity.

	(1)	(2)	(3)	(4)
	Connecting to			
	any city in admission province		the capital of admission province	
connect	0.0590	0.440***	−0.0272	0.297***
	(0.0824)	(0.0830)	(0.0858)	(0.0910)
College-admission province-track-batch FE	✓	✓	✓	✓
Admission province-track-batch-year FE	✓	✓	✓	✓
College-track-batch-year FE	✓	✓	✓	✓
N	417,516	417,516	417,516	417,516
R-squared	0.976	0.976	0.976	0.976

Note: In Columns (1) and (3), the bilateral HSR connectivity variable *connect* is defined as the college city and the admission province being in the same HSR network. In Columns (2) and (4), variable *connect* is defined as the college city and the admission province being on the same HSR line. The values in parentheses represent college city-admission province level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

To address this, we consider bilateral HSR connectivity on the same HSR line: if the college city and any city in the admission province are located on the same HSR line, there is an HSR connection between the college city and the admission province. Column (2) of Table 2 reports the results. We find that when the college city and admission province are connected by the same HSR line, the college's admission scores in the admission province increase significantly by 0.44 percentile points. This suggests that the significant positive effect of bilateral HSR connectivity on admission scores is primarily driven by connections on the same HSR line rather than by being in the same HSR network.

In Columns (3) and (4) of Table 2, we construct a new bilateral HSR connectivity measure using the college city and the capitals of admission province. The results in Columns (3) and (4) are consistent: HSR connectivity between the college city and admission province significantly increases the college's admission scores, and this significant positive effect is primarily driven by connections on the same HSR line rather than by being in the same HSR network.

One might raise the following question: the bilateral HSR connection effect reported in Table 2 appears to be concentrated along the same HSR line, which may partly reflect distance heterogeneity. To address this question, we introduce bilateral distance interactions based on Table 2. Specifically, we interact the bilateral HSR connectivity variable with the logarithm of the distance between the city where the college is located and the capital of the admissions province. Appendix Table D2 reports the results of this interaction analysis. We find significant distance heterogeneity in the bilateral connectivity effect: the greater the distance between the city where the college is located and the admissions province, the weaker the bilateral HSR connectivity effect. The distance heterogeneity results reported in Appendix Table D2 are consistent with the predictions of the gravity model and further corroborate the findings in Table 2.

Based on the above analyses, we conclude that HSR connectivity between college city and admission province—particularly when they are on the same HSR line—generates a significant transportation convenience improvement. The HSR reduces bilateral commuting time and costs, enhancing candidates' willingness to apply, which in turn significantly raises the college's admission scores in the admission province.

6. Further analysis

6.1. The role of bilateral distance

In this section, we exploit the bilateral structure of our data to examine the role of distance. In Table 3, we construct three indicators to measure the relative location between colleges and the admission provinces, and interact these with the HSR opening variable *hsr*. The first indicator, *inner*, equals 1 if the college is located within the admission provinces, and 0 otherwise. The second indicator, *contig*, equals 1 if the college's province shares a border with the admission provinces, and 0 otherwise. The third indicator, *1(distance ≤ 1000)*, equals 1 if the distance between the college's city and the capital of the admission provinces is within 1000 km, and 0 otherwise. The results in Table 3 show that the coefficients of all three interaction terms are significantly positive. This suggests that the closer the bilateral distance, the stronger the positive effect of HSR on average admission scores.

To provide further evidence on the role of bilateral distance, we divide the distance between colleges and admission provinces into concentric rings of 200-km intervals and construct corresponding dummy variables. This process continues up to the 1200–1400 km ring, with all distances beyond 1400 km grouped into the final category. We then interact each of these distance ring dummies with the core independent variable from Equation (2), and include all interaction terms in a single regression. Fig. 5 presents the results. It shows that within 800 km, the effect of HSR on average admission scores is strong. Between 800 and 1200 km, the effect attenuates but remains statistically significant. When the bilateral distance exceeds 1200 km, the effect becomes small and statistically insignificant. These findings are consistent with existing estimates in the literature regarding the effective transport range of China's HSR system. Due to differences in methods and data, the estimated optimal HSR transport distance varies across studies, with some suggesting around 1000 km (Zhang et al., 2017, 2018) and 700 km (Chai et al., 2018).

6.2. The role of bilateral development gap

In this section, we investigate the heterogeneous effects related to regional development gap within China. We exploit the directional nature of our data: candidates located in admission province p choose to apply to college i located in city c, as illustrated in Equations (1) and (2).

Table 3
The role of bilateral distance.

	(1)	(2)	(3)
hsr	0.328***	0.274**	0.0353
	(0.111)	(0.119)	(0.127)
inner × *hsr*	0.771***		
	(0.183)		
contig × *hsr*		0.652***	
		(0.128)	
1 (distance ≤ 1000) × *hsr*			0.722***
			(0.119)
Controls	✓	✓	✓
College-admission province-track-batch FE	✓	✓	✓
Year-admission province-track-batch FE	✓	✓	✓
N	416,124	391532	416,124
R-squared	0.969	0.969	0.969

Note: The dependent variable is the average admission score aggregated at the college–admission province–track–batch–year level. Variables *inner*, *contig*, and 1 (*distance* ≤ 1000) represent: (i) the college is located within the admission provinces; (ii) the college's province shares a border with the admission provinces; and (iii) the distance between the college's city and the capital of the admission provinces is within 1000 km, respectively. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

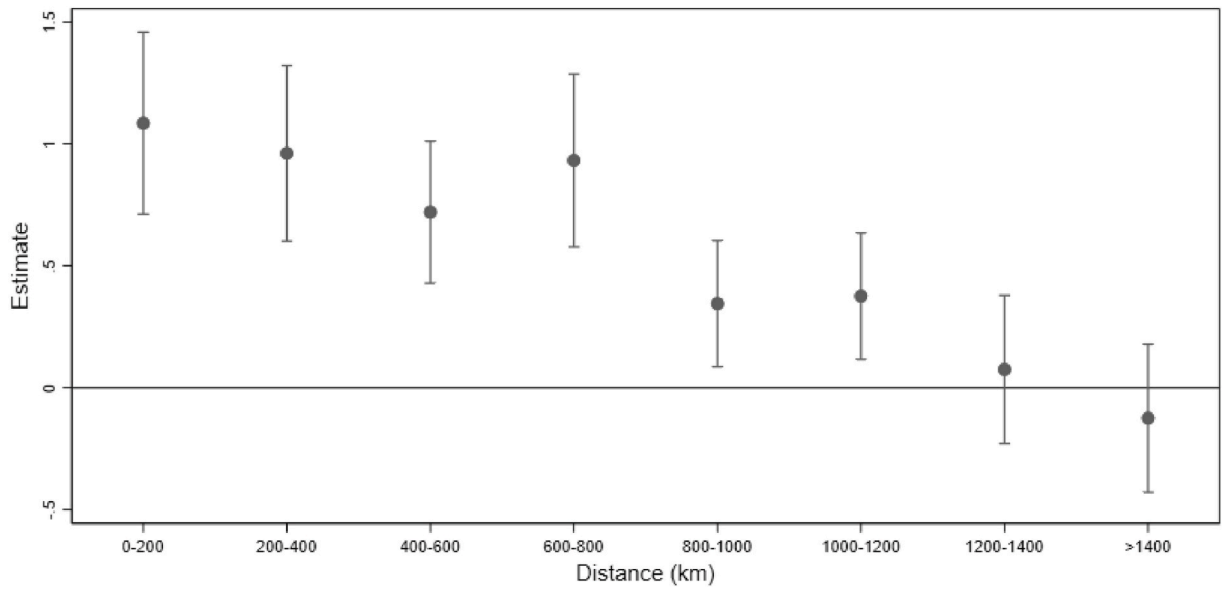

Fig. 5. The core effect by bilateral distance category.
Note: Fig. 5 shows the estimated coefficients and 95% confidence intervals.

To measure regional development gap, we use per capita GDP. Specifically, we define a development gap indicator, *gap*1, as the difference in the logarithm of per capita GDP between the college city and the admission province. This variable reflects the economic development gap between the college city and the student's hometown at the time of application. We interact *gap*1 with the core independent variable *hsr*, and report the regression results in Column (1) of Table 4. The coefficient on the interaction term *gap*1 × *hsr* is positive and statistically significant, implying that the more economically developed the college city is relative to the student's home province, the stronger the positive effect of HSR on the average admission scores. This suggests that the HSR network has exacerbated disparities in college admissions quality across regions in China: colleges located in economically developed areas are better able to attract higher-quality students through the HSR network. To test the robustness of this finding, we construct alternative gap indicators *gap*2 and *gap*3 using the GDP of the capital city of the admission province and the per capita GDP of the most developed city in the admission province, respectively. Columns (2) and (3) of Table 4 present these regression results. Furthermore, in Appendix Table E1, we use per capita GDP at the provincial level for the college location instead of city-level GDP to construct the gap indicator. The results in Table 4 Columns (2) and (3), as well as those in Appendix Table E1, all yield qualitatively similar conclusions. Additionally, in Appendix D, we discuss the potential reasons for the differences in the estimated coefficients of the gap indicators between Table 4 and Appendix Table E1.

Table 4
The heterogeneous effects related to regional development gap.

	(1)	(2)	(3)
hsr	0.211	0.418***	0.527***
	(0.152)	(0.107)	(0.111)
gap1 × *hsr*	0.433**		
	(0.200)		
gap1	−1.330***		
	(0.474)		
gap2 × *hsr*		0.449**	
		(0.209)	
gap2		−1.290***	
		(0.466)	
gap3 × *hsr*			0.435**
			(0.186)
gap3			−1.275***
			(0.461)
Controls	✓	✓	✓
College-admission province-track-batch FE	✓	✓	✓
Year-admission province-track-batch FE	✓	✓	✓
N	416,124	415,945	415,945
R-squared	0.969	0.969	0.969

Note: The dependent variable is the average admission score aggregated at the college–admission province–track–batch–year level. Variables *gap*1, *gap*2, and *gap*3 represent: (i) the difference in the logarithm of per capita GDP between the college city and the admission province; (ii) the difference in the logarithm of per capita GDP between the college city and the capital city of the admission province; and (iii) the difference in the logarithm of per capita GDP between the college city and the per capita GDP of the most developed city in the admission province, respectively. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

To provide further evidence, we leverage the geographic distribution of economic development in China. China's economy is unevenly distributed across regions: the eastern region is significantly more developed than the western region. Following the classification standard of the National Bureau of Statistics of China,[11] we divide provinces into eastern and western regions. Utilizing the directional nature of our dataset, we define four dummy variables to capture student-college flows: (i) students from the eastern region applying to colleges in the western region; (ii) students from the western region applying to colleges in the eastern region; (iii) students from the eastern region applying to colleges within the eastern region; and (iv) students from the western region applying to colleges within the western region. We interact each of these dummy variables with the core independent variable *hsr* from Equation (2) and include all interaction terms in a single regression.

Fig. 6 presents the results. We find that the core effect of HSR is primarily driven by flows from the western region to eastern colleges, and by within-eastern flows. In contrast, the effect is small and statistically insignificant for flows from the east to the west and within the western region. This indicates that the HSR-induced improvement in college admissions quality arises only in two types of flows: from developing to developed regions, and within developed regions. In

[11] https://www.stats.gov.cn/hd/cjwtjd/202302/t20230207_1902279.html.

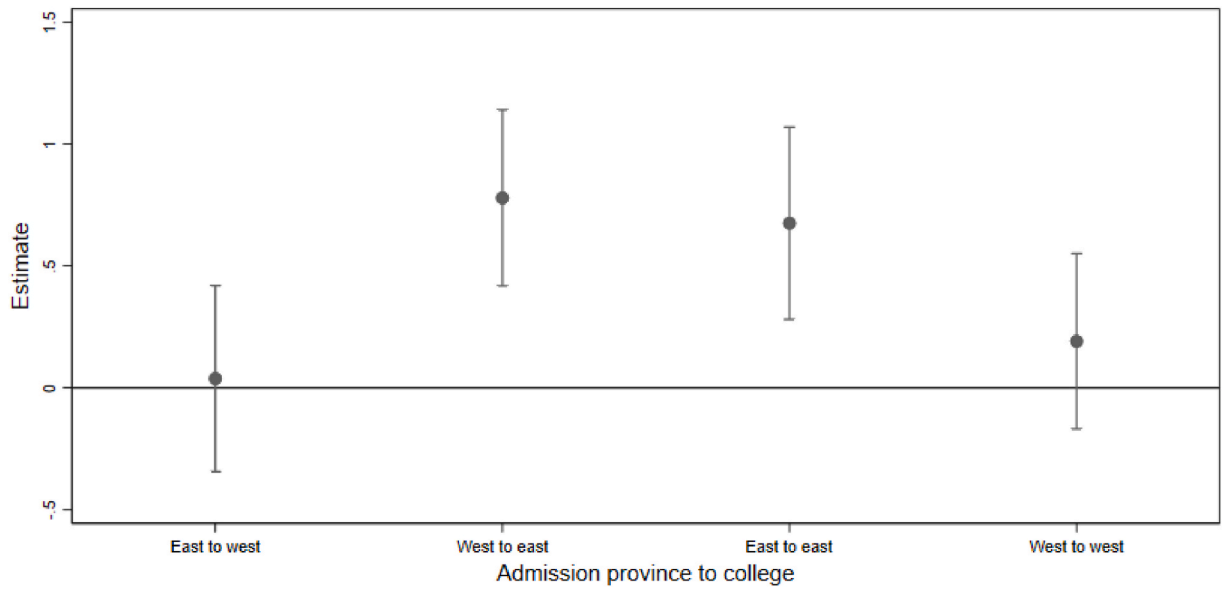

Fig. 6. The core effect by directional nature (eastern region and western region).
Note: Fig. 6 shows the estimated coefficients and 95% confidence intervals. The "East" refers to the eastern area as defined by the classification of the National Bureau of Statistics of China. The "West" includes all areas not classified as part of the eastern region under this standard.

contrast, flows from developed to developing areas, or within developing regions, do not benefit from HSR. We conclude that colleges in more developed regions are able to attract higher-quality students from developing areas via the HSR network, whereas colleges in developing regions do not enjoy comparable benefits.

One might raise the question of whether HSR-induced outside options affects the admission scores of local students, and whether this effect differs between developed and developing regions. To address this, we conduct an empirical analysis. It should be noted that, due to the lack of candidate-level data, we cannot directly observe local students' decision-making. We adopt an indirect approach by restricting the college admission score sample to local admissions, that is, cases where the college's province and the admissions province are the same. This restricted sample contains information only on local students applying to local colleges. To capture the outside options induced by HSR, we construct four binary HSR connectivity variables at the admissions-province level: eastern provinces connected to western regions, western provinces connected to eastern regions, eastern provinces connected to eastern regions, and western provinces connected to western regions. A detailed description of the variables and the econometric model, as well as a comprehensive analysis of the regression results, is provided in Appendix E. The regression results in Appendix Table E2 show no evidence that HSR-induced outside options has a significant effect on the admission scores of local students. Additionally, this effect does not appear to differ significantly between admissions provinces in developed and developing regions. It should be noted, however, that the analysis in Appendix Table E2 has two limitations. First, due to the inability to directly observe local students' decision-making, the results provide only an indirect assessment. Second, the HSR connectivity indicators measured at the provincial level are relatively coarse and may not fully capture the outside options induced by HSR connectivity at the provincial level. Therefore, we are inclined to treat the results in Appendix Table E2 as suggestive evidence.

6.3. Supplementary empirical analysis based on population census data

In the preceding analysis, our empirical results show that the opening of HSR significantly improves college admission quality, as reflected in higher admission scores. One may question whether this effect merely indicates an improvement in admission quality, without shedding light on whether colleges enroll a larger number of students. It is important to clarify that the college admission score data used in this study are obtained from Sina Gaokao. However, Sina Gaokao does not provide information on planned enrollment quotas or actual enrollment numbers of colleges. Consequently, we are unable to examine, based on admission score data, whether HSR induces colleges to admit more students.

To address this issue, we turn to data from the Chinese population censuses. The Chinese population census is conducted every five years. Years ending in zero correspond to the full population census (commonly referred to as the "major census"), while years ending in five correspond to the 1 percent population sample survey (the "mini census"). The Chinese population census is well known for its comprehensive coverage and high data quality and has been widely used in academic research. Even the "mini census" exhibits strong representativeness (Huang and Dong, 2025; Liu et al., 2025).

Given the study period of the college admission score data (2006–2018) and data availability,[12] we utilize the 2005 and 2015 population censuses. These censuses provide rich individual-level information, including respondents' place of household registration (hukou), place of enumeration, and educational attainment (primary school, junior high school, senior high school, junior college, undergraduate, and postgraduate). The hukou system is a region-based household registration unique to China. Since the 1980s, China has gradually relaxed restrictions on population mobility (Wang et al., 2021), giving rise to one of the largest population and labor movements in the world—China's "Great Migration" (Facchini et al., 2019; Tombe and Zhu, 2019; Ma and Tang, 2020). In the twenty-first century, while hukou status remains linked to access to local public services and social welfare, it generally does not restrict interregional mobility (Chen et al., 2026).

In a given census wave, if a respondent's place of hukou registration differs from the place of enumeration, we interpret this as migration from the hukou location to the survey location. More importantly, the 2005 and 2015 censuses also report individuals' educational attainment status (enrolled, graduated, completed but not graduated, or dropped out). Using information on currently enrolled students, we are able to

[12] To the best of our knowledge, data from the 2020 population census are not yet publicly available.

observe college attendance that involves migration from the hukou location to the place of enumeration. This allows us to complement the analysis based on college admission scores by directly examining student flows associated with higher education enrollment.

Using data from the 2005 and 2015 population censuses, we compute the scale of population flows between pairs of Chinese cities. To examine the impact of HSR on intercity migration of highly educated individuals and enrolled students, we construct a gravity-style long-difference model, as specified in Equation (5):

$$\Delta hflow_{ij}^{i\neq j} = \alpha + \beta \Delta connect_{ij} + \mathbf{\Delta X_{ij}^{i\neq j}} + \gamma_i + \delta_j + \varepsilon_{ij} \quad (5)$$

Here, i denotes the hukou city and j the city of enumeration. The variable $hflow_{ij}^{i\neq j}$ measures the size of the flow of highly educated individuals migrating from hukou city i to city j. The variable $connect_{ij}$ captures the HSR connectivity between hukou city i and city j. Following Section 5.3, we define an HSR connection between cities i and j as an indicator equal to one if the two cities are located on the same HSR line, and zero otherwise. The operator Δ denotes the difference between 2005 and 2015. Specifically, $\Delta hflow_{ij}^{i\neq j} = hflow_{ij}^{2015} - hflow_{ij}^{2005}$ is the long difference in the flow of highly educated individuals between city pair ij, and $\Delta connect_{ij} = connect_{ij}^{2015} - connect_{ij}^{2005}$ is the corresponding change in HSR connectivity. $\mathbf{X_{ij}^{i\neq j}}$ is a vector of control variables. To control for overall migration intensity between cities, we include measures of total population flows and flows of the educated population (primary education and above) between city pairs. All control variables are also expressed in long differences. γ_i denotes hukou city fixed effects, and δ_j denotes destination city fixed effects. These fixed effects absorb the push–pull factors in the gravity framework. All bilateral time-invariant factors (such as geographic distance and cultural differences) are eliminated by the differencing operator Δ. ε_{ij} is an idiosyncratic error term. Robust standard errors are clustered at the city-pair (city i–city j) level. Descriptive statistics for the main variables are reported in Appendix Table E3.

To facilitate an intuitive interpretation of the economic magnitude of the core effect, we apply a normalization to the variable $\Delta hflow_{ij}^{i\neq j}$ similar to that in Equation (1). The estimated coefficient $\widehat{\beta}$ is the parameter of primary interest. It implies that HSR connectivity between two cities leads to a change of $\widehat{\beta}$ percentage points in the percentile rank of intercity flows of highly educated individuals.

To align with the analysis based on college admission scores data, we focus on enrolled undergraduate students as the relevant highly educated group. Column (1) of Table 5 reports the estimation results. We find that HSR connectivity between cities significantly facilitates intercity flows of the enrolled undergraduate population. Specifically, the results in column (1) indicate that, following HSR connection, the percentile rank of the flow of enrolled undergraduate population from city i to city j increases by 0.86 percentage points. In column (2), we turn to enrolled postgraduate students as another group of highly educated individuals. The estimates in column (2) show that HSR connectivity also significantly promotes intercity migration of enrolled postgraduate population. In column (3), we pool undergraduate and postgraduate, and the qualitative conclusions remain similar.

Finally, we examine whether the effects identified in this section exhibit regional heterogeneity. Following the analysis in Section 6.2, we focus on whether highly educated individuals from the western region tend to migrate after HSR connections to the eastern region. We adopt an approach similar to that used in the college admission score analysis, defining four dummy variables: (i) flows from eastern cities to western cities, (ii) flows from western cities to eastern cities, (iii) flows between eastern cities, and (iv) flows between western cities. Each of these dummy variables is interacted with the independent variable $\Delta connect_{ij}$ from Equation (5), and all interaction terms are included in the same regression. For ease of comparison, the regression results are plotted in Fig. 7, with the full estimates reported in Appendix Table E4. Fig. 7 and Appendix Table E4 show that HSR connectivity significantly promotes flows of enrolled undergraduate population within the eastern region, within the western region, and from the western region to the eastern region. In contrast, the effect of HSR connectivity on flows of undergraduates from the eastern region to the western region is small or statistically insignificant. The qualitative conclusions for postgraduate students are analogous.

These empirical findings are consistent with those in Section 6.2 based on college admission scores, indicating that the HSR has widened regional disparities in China. Our analysis shows that when the western region becomes connected to the eastern region via HSR, the number of students from the western region attending colleges in the eastern region increase significantly. In contrast, when the eastern region is connected to the western region by HSR, there is no comparable increase in the magnitude or significance of flows. This implies that, relative to the eastern region, HSR has induced a net outflow of talent from the western region. Importantly, this core finding is consistently supported by empirical analyses based on both college admission score data and population census data.

Table 5
Long-difference regression results based on population census data.

	(1)	(2)	(3)
	Enrolled		
	Undergraduate	Postgraduate	Undergraduate + Postgraduate
$\Delta connect_{ij}$	0.860***	0.193***	0.888***
	(0.0898)	(0.0434)	(0.0905)
Controls	✓	✓	✓
City i FE	✓	✓	✓
City j FE	✓	✓	✓
N	87,912	87,912	87,912
R-squared	0.102	0.064	0.108

Note: The dependent variable measures the size of the highly educated population flowing from city i to city j. The dependent variable includes only currently enrolled students. Variable $connect_{ij}$ is defined as the city i and city j being on the same HSR line. The operator Δ denotes the difference between 2005 and 2015. The values in parentheses represent city-pair (city i–city j) level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

7. Conclusion remarks

In this paper, we examine the impact of transportation infrastructure on the quality of college admissions. We exploit the unique institutional background of China's NCEE system and the rapid expansion of its HSR network. Using a DID approach and detailed data on college admission cut-off scores data, we provide empirical evidence that the opening of HSR significantly improves the quality of college admissions, as reflected in both higher average and maximum admission scores. We explore three potential mechanisms: (1) HSR promotes city economic development, (2) HSR strengthens college capacity, and (3) HSR improves transportation accessibility. By sequentially incorporating relevant mechanism variables into the baseline regression, we find that the first two mechanisms cannot account for the core effect. For the third mechanism, we provide suggestive evidence that the positive impact of HSR on admission quality is stronger in cities with more central positions in the HSR network. In addition, our empirical analysis finds that bilateral HSR connectivity significantly increases college admission scores. We find that bilateral geographic distance weakens the positive effect of HSR, and our estimates of this distance effect are consistent with existing literature on the optimal transport range of China's HSR system. In addition, we present evidence suggesting that HSR may exacerbate regional inequality in college admissions quality: colleges in

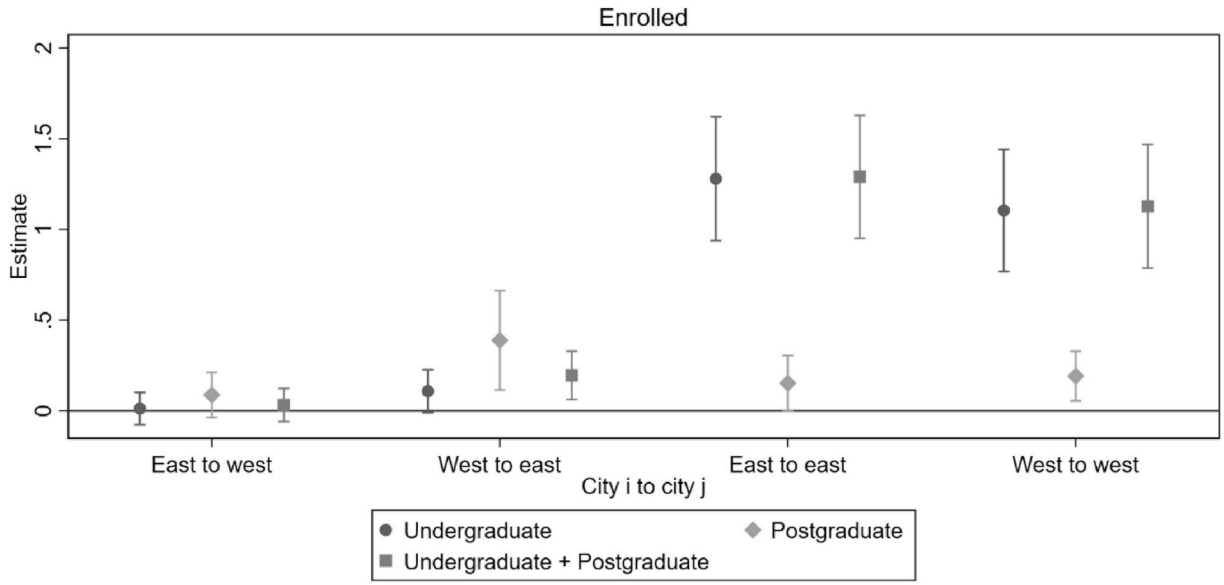

Fig. 7. Regional heterogeneity in the impact of bilateral HSR connectivity on the mobility of highly educated individuals (enrolled).
Note: Fig. 7 shows the estimated coefficients and 95% confidence intervals. The "East" refers to the eastern area as defined by the classification of the National Bureau of Statistics of China. The "West" includes all areas not classified as part of the eastern region under this standard. The dependent variable includes only currently enrolled students.

economically developed regions are able to attract higher-quality students from developing areas via HSR, whereas institutions in developing regions do not experience comparable benefits. Finally, based on Chinese population census data, our empirical analysis finds that bilateral HSR connectivity significantly promotes the mobility of highly educated enrolled students. This effect exhibits notable regional heterogeneity: compared with developing regions, developed regions experience a disproportionately higher inflow of talent.

This paper contributes to three strands of the literature. The first relates to transportation infrastructure, particularly the literature on HSR. The second focuses on NCEE. The third concerns population and labor mobility, with an emphasis on the movement of talent. To the best of our knowledge, this is the first study to investigate the impact of HSR on college using data on admission scores data. This paper makes three main contributions. First, we provide empirical evidence that HSR not only enhances college admissions quality but also amplifies regional inequalities in admission outcomes. Second, our findings highlight that the external environment of colleges plays a critical role in students' application decisions. Third, we offer new empirical evidence on how HSR affects talent mobility and the spatial distribution of talent in China.

CRediT authorship contribution statement

Tianyu Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Lei Li:** Writing – review & editing, Project administration, Conceptualization. **You Lu:** Writing – original draft, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.regsciurbeco.2026.104223.

Data availability

Data will be made available on request.

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